

Request for Proposal

Metric Pre-Flight Diagnostic System for
Professional FPV Drone Operations

Engineering Science

ESC102: Praxis II

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Abstract

A single equipment failure can expose first-person view (FPV) drone videographers to liability of up to \$12.5 million [Appendix A, 2.1]. Through interviews with FPV drone pilots/engineers and secondary research, it was revealed that pre-flight preparations rely heavily on intuition to determine the drone's health [Appendix A, 2.5]. In the event of a lawsuit, the plaintiff may question the safety of the drone's operation due to their reliance on intuition. This RFP outlines the opportunity of providing metric, pre-flight data to evaluate the integrity of FPV drone equipment, such as motors and batteries. This opportunity is appropriately scoped for students to design and prototype within the six-week time constraint, while maintaining meaningful engagement with the community.

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1 Introduction

The purpose of this Request for Proposal is to introduce the community of professional First Person View (FPV) videographers in the GTA and to recommend the opportunity of metric indicators for pre-flight drone diagnostic checks. As FPV cinematography transitions from a hobbyist pursuit to a high-stakes professional service, it has emerged as a cost-effective alternative to traditional aerial filming methods, such as helicopters or vehicle-mounted rigs. However, while these drones offer reduced setup times and smaller crew sizes, they operate under high-risk situations. FPV videographers often pilot custom-built aircraft in high-density urban environments that can have legal consequences if the drone fails mid-flight. The industry currently lacks a standardized method for equipment health verification, leaving a critical gap in safety protocols. Professional operators currently rely on intuition, such as "audio checks," to determine if a drone is safe for flight [Appendix A, 2.5].

In an industry where a mechanical failure can result in catastrophic legal and physical consequences, there is an urgent need for centralized, objective evaluation methods to replace these arbitrary and high-risk decision-making processes. This RFP will give an overview of the community and their lived experience, and then introduce the primary and secondary stakeholders. The document will then state a key problem framed by the community's lived experience and discuss multiple feasible solutions within constraints proposed by stakeholders. Subsequently, the RFP will cover the expectations of the final solution's function and provide a brief on the ethics of the solution.

1.1 Community Overview

The drone-based videography community in the Greater Toronto Area is made up of enthusiastic and passionate individuals with interests in multiple intersecting fields, from the technicalities of drone piloting to the art of photography and videography. This community, while encompassing a diverse array of interests, maintains a close network between its members [Appendix A, 1.1, 1.2]. For one, there are a variety of social media groups on platforms like Facebook for this community to share ideas, information, and plan in-person meetups [1]. There also exist potential points of interaction in the form of FPV (first-person video) drone racing meetups and networking events for general UAV (unmanned aerial vehicle) groups [2] [3]. In this context, FPV refers to a piloting method where the operator controls the drone via an immersive video feed transmitted to goggles, while UAV serves as the broader industry classification for any aircraft operated without a pilot on board.

One professional forum for this community is D.Vision Aerials, a Toronto-based company providing FPV drone services, typically for brand marketing and film [4]. This group emphasizes the more technical and commercial side of the drone videography community, showing how the skills amassed by individual hobbyists, such as racing and piloting and/or an eye for camera aesthetics, can translate into valuable expertise in jobs [Appendix A, 1.3]. Ultimately, companies like D.Vision Aerials demonstrate the bridge between the individual hobby and performance culture of drone filming to the professional, full-time side.

1.2 Lived Experience

The founder of D.Vision Aerials, Danny Nguyen, explicitly stated the differences between hobby and professional FPV drone piloting. Unlike hobbyists, Nguyen's team operates under strict federal law and a \$12.5 million liability policy [Appendix A, 2.1], meaning every flight carries significant professional risk. To manage this, the team maintains a massive fleet of approximately 130 custom-built drones and orders "batches of 30 to 50" battery sets annually to sustain their operations [Appendix A, 2.6]. Nguyen described his typical shoot day as intense and high tempo, stating that the team must be ready to launch within 30 seconds of a director's call to avoid delaying production [Appendix A, 2.11], which introduces a strict time constraint.

Meanwhile, Nguyen recounted his "out-of-body" experience while flying the drones [Appendix A, 2.2], highlighting how immersive the piloting experience can be and the attention required in a typical shooting job. As a result of the strict demands for time and attention, Nguyen described the pre-flight checks as subjective and unreliable, where he tests the motors by spinning them up and relying on auditory signals to determine the motor's health. Similarly, managing their inventory of batteries is described as a "guessing game" due to poor manufacturing quality control; Nguyen noted that some packs fail within six months while others last years, with no objective way to distinguish them before a flight [Appendix A, 2.6].

1.3 The Current Process

Based on consultations with GTA community leads [Appendix A, 2.1], the following enumerated list details the typical operational workflow from the moment a crew arrives on set to the moment of take-off. This process highlights the reliance on sensory intuition and the critical 30-second bottleneck.

1. Radio Frequency Selection:

Upon arrival, the pilot and Operations Officer scan the environment for open radio frequencies to connect their drones to. However, in dense urban environments like Downtown Toronto, "invisible" threats such as high-power Wi-Fi routers or smart devices can trigger catastrophic signal loss [Appendix A, 2.7].

2. Physical Inspection:

The pilot performs a manual "flex test" on the drone's carbon fibre frame. They apply physical pressure to the arms to feel for micro-fractures or delamination that may have occurred during transport. This is a purely qualitative assessment based on the pilot's experience with the specific aircraft's rigidity.

3. Batteries:

The engineer selects a battery from a fleet of dozens. Because the community uses "dumb" batteries without internal health tracking, this selection is often based on handwritten labels or "gut feeling" about which packs are newer. The pilot uses a basic voltage checker to ensure the pack is "full," but has no way to verify if the internal resistance has spiked due to cold weather exposure

(-15°C) until the drone is already in the air [Appendix A, 2.6 & 2.13].

4. **Subjective Motor Verification:**

Before flying, the pilot brings the drone close to their ear to listen for specific sound defects. If the motor "sounds smooth," it is cleared for flight. This is the primary point of failure in the current safety protocol, as human hearing cannot detect high-frequency bearing harmonics or internal winding fatigue [Appendix A, 2.5].

5. **The 30-Second Setup Window:**

The drone team is given a window, often as short as 30 seconds, to "Plug and Go" [Appendix A, 2.11]. Under this intense time pressure, the pilot must decide to commit to the flight based on the subjective data gathered in the previous steps.

6. **High-Velocity Environment:**

The drone is launched, often reaching speeds of 180 km/h [Appendix A, 2.16]. Any error in the previous six steps, a missed motor rattle or a "sagging" battery may result in a crash.

2 Stakeholders

Primary Stakeholders: The FPV Cinematography Team

1. The FPV Drone Pilot

- **Role & Context:** The pilot operates the drones and uses their specialized flight skills to minimize damage to the equipment, the environment, and surrounding humans. They are responsible for the high-speed execution of cinematic shots in complex environments, often flying custom drones at speeds exceeding 180 km/h [Appendix A, 2.16].
- **Lived Experience:** While they operate in a high-intensity, high-stakes environment, the pilot maintains a "chill and relaxed" mood during flights, relying on deep trust in their own abilities and the machine [Appendix A, 2.4].
- **Impact of Design:** Their ability to operate is directly influenced by the drone's physics. Any potential design solution mounted to the drone must not negatively impact the center of mass or power-to-weight ratio, as "if you add like 10 grams, it's not gonna fly the right way" [Appendix A, 2.10].
- **Key Values:**
 - Trust in the equipment's reliability
 - Agility in flight performance
 - Simplicity in the user interface

2. The Drone Engineer/Builder

- **Role & Context:** The engineer is responsible for the design, assembly, and maintenance of the company's ~130 custom-built drones to ensure optimal performance in the air. They are the primary user of any diagnostic or maintenance tools.
- **Lived Experience:** They are responsible for conducting safety checks and maintaining the drones' operability. Currently, this involves subjective methods (like listening to motors) [Appendix A, 2.5] or using battery checking devices to detect suboptimal battery capacities [Appendix A, 2.6].
- **Key Values:**
 - Certainty in hardware status (validating "Go/No-Go")
 - Efficiency in the maintenance workflow
 - Longevity/reliability of the equipment components

3. The Operations & Compliance Officer

- **Role & Context:** The operations officer is responsible for complying with Transport Canada regulations and managing the SFOCs (Special Flight Operations Certificates) and permits.
- **Lived Experience:** They ensure the team operates legally within the demands of their clients, acting as the bridge between strict federal regulations, which changed recently last year [Appendix A, 2.15], and the chaotic reality of a film set.
- **Key Values:**
 - Legitimacy of operations
 - Accountability via documentation/logs
 - Risk Avoidance regarding legal penalties

Secondary Stakeholders:

1. The Production Client (Producers/Directors)

- **Role & Context:** The leader of the project who enforces strict schedules. They view the drone team as a specific tool to achieve an artistic vision.
- **Lived Experience:** They value time above all else, managing budgets where even standard short film production costs typically range between \$700 and \$1,500 per minute of screen time [5]. Consequently, any technical delay from the drone team, such as exceeding their standard 30-second reset window [Appendix A, 2.11], directly impacts the project's budget.
- **Key Values:**
 - Speed of workflow (Zero Downtime)
 - Predictability of results
 - Cost-Effectiveness

2. On-Set Actors & Bystanders

- **Role & Context:** The humans (actors, drivers, crowds) in physical proximity to the high-speed drones. They are the passive recipients of risk.
- **Lived Experience:** Their physical safety depends entirely on the reliability of the drones and the competence of the pilot. This safety margin could be compromised by designs that modify the operability of the drones. They represent the high stakes of failure.
- **Key Values:**
 - Personal Safety
 - Trust in the production team's competence.

Tertiary Stakeholders:

1. Transport Canada

- **Role & Context:** The federal regulator that requires pilots to obtain permits (SFOCs) prior to their planned flight to ensure safety within the area [Appendix A, 2.14]. They ensure proper use of the airspace and manage availability in requested locations.
- **Impact:** They grant the license to operate. Without their approval, the primary community cannot legally function.
- **Key Values:**
 - Public Safety
 - Adherence to Regulation
 - Airspace Integrity

2. Liability Insurance Companies

- **Role & Context:** They cover the \$12.5M liability policy required for professional operation [Appendix A, 2.1].
- **Impact:** They are the financial safety net. If accident rates rise, premiums rise, potentially making the business model unsustainable.
- **Key Values:**
 - Risk Reduction
 - Quantifiable Safety Measures
 - Asset Protection

3 Team Positionality and Bias

3.1 Our Position in Context

The authoring team approaches this opportunity from the perspective of first-year Engineering Science students. While we possess foundational knowledge in physics, programming, and mechanical design, our experience is primarily academic and structured. This stands in contrast to the Primary Stakeholders (FPV Pilots), who operate in a high-stakes commercial environment where "gut feeling," sensory intuition, and rapid execution are prioritized over formal engineering documentation. Our position as external observers allows us to identify inefficiencies (lack of data logs) that the community has normalized, but it also separates us from the physical and financial risks they face daily.

3.2 Identified Biases

Our position introduces three specific biases that influenced our interpretation of the community's lived experience:

1. **Objectivity vs. Lived Experience:** Based on our experience as engineering students, we are predisposed to value clear, quantitative metrics over methods with greater human involvement. This causes us to risk overvaluing efficiency while neglecting the human base behind the community and the hands-on skills brought by drone pilots.
2. **Safety vs. Speed:** Given our adherence to safety and robustness in past designs, we are inclined to place these qualities above others, like time efficiency, when approaching this opportunity. This may cause us to underestimate the needs of stakeholders for speed and portability due to the redundancy of in-depth safety checks.
3. **Complexity vs. Practicality:** In approaching this community, we also expected more technically-based opportunities involving complex mechanisms and hardware fixes. This could potentially sway us away from simpler issues that may ultimately be more critical for members of the community.

3.3 Mitigation Strategies

To mitigate these biases and ensure the RFP reflects the community's reality rather than our assumptions, we employed the following strategies:

- Throughout contacting primary sources, we prioritized open-ended prompts ("Walk us through your pre-flight routine") over solution-focused questions to more broadly prevent our preconceived notions from impacting the opportunities we would identify. This allowed the "Black Box" maintenance problem to emerge naturally from the stakeholder's narrative rather than being imposed by us.
- Additionally, in interviewing, to check our bias toward complex/heavy engineering solutions, we explicitly asked for "deal-breakers". The stakeholder's strict constraints (<10g added weight, <30-second setup) forced us to pivot away from over-engineered concepts to those fitting their agile

workflow.

- When landing on an opportunity, to counteract our bias toward objectivity, it was framed not as replacing the intuition and skill of pilots, but as supplementing their lived experience. We primarily seek to provide pilots with a tool that can more objectively validate their instincts and prevent expensive damage.

4 The Opportunity - Objective Testing of Hardware

The Problem: The Subjectivity Gap in Pre-Flight Maintenance

Currently, certain core aspects of pre-flight maintenance on FPV drones lack objective, standardized methods of evaluation. The community relies on subjectivity, a dangerous reliance on sensory intuition to determine airworthiness. For example, verifying motor integrity often amounts to "audio hearing," in which an operator listens for a specific pitch or rattle to identify internal damage [Appendix A, 2.5]. This method is inherently unreliable as it cannot detect sub-threshold failure modes, such as micro-fractures in carbon fibre frames or invisible winding fatigue in brushless DC motors.

Similarly, while the community utilizes basic voltage checkers, there is no precise method for quality control tailored to the rapid, high-load requirements of professional cinematography. Battery health is treated as a guessing game, where operators struggle to distinguish between a healthy high-discharge cell and a degraded one that may sag under the stress of a high-speed shoot [Appendix A, 2.6].

Justification 1: High Stakes and Professional Liability

The stakes for these subjective checks are extreme. Professional FPV videographers operate in high-density urban areas, often flying at speeds exceeding 180 km/h in close proximity to actors, crowds, and high-value equipment (see Section II). Given that the community is legally bound by \$12.5 million liability policies [Appendix A, 2.1], the current reliance on "gut instinct" represents a significant professional and legal vulnerability. A single undetected mechanical flaw can result in catastrophic property damage or personal injury, potentially ending an operator's career.

Justification 2: Environmental Stress and Temporal Constraints

The hostile environments of the GTA exacerbate this problem. Frequent exposure to -15°C winters significantly alters battery chemistry and conductivity [Appendix A, 2.13], while the use of waterproofing agents can trap heat and accelerate the degradation of electronics. Compounding this is the "Time is Money" pressure of a film set; to avoid costly production delays, operators must be ready to launch within a 30-second window [Appendix A, 2.11]. This pressure often forces operators to prioritize speed over thorough, albeit manual, safety verifications.

The Opportunity: Objective, Field-Ready Diagnostics

There is a clear engineering opportunity to design a portable diagnostic system that provides the

community with clear, objective feedback on a drone's electromechanical health. This solution must bridge the gap between speed and certainty, transforming pre-flight maintenance from a subjective ritual into a validated engineering protocol.

5 Opportunity Needs, Goals, and Objectives



Figure 1: Flowchart depicting needs, goals, and objectives for addressing the opportunity.

To successfully transition the FPV community from subjective "audio checks" to a professional engineering protocol, any proposed solution must satisfy the following technical and operational criteria.

Need: A reliable, diagnostic system to verify the integrity of the drone's systems, evaluate the flight-readiness of its core components, and prevent in-flight failure. A unified, field-deployable flight-readiness verification system that objectively validates its core components to prevent in-flight failure, integrates seamlessly into high-tempo professional workflows, operates reliably in hostile environmental conditions, and ensures regulatory accountability through data logging.

- **Goal 1: Quantify motor soundness and integrity before flight.**
 - **Objective 1.1:** The system shall provide a definitive assessment of motor health that identifies mechanical irregularities undetectable by human audio checks.
 - * **Justification:** The existing practice of listening to motors is unreliable, especially in the midst of a busy film set that can add background noise and stress. Some motor damage may also be minor enough not to be noticed prior to flight but still lead to drone failure in

action [6]. Any system addressing the opportunity must eliminate the need for guesswork and provide data-backed assurance of safety.

- **Objective 1.2:** The system shall distinguish between easily addressable external issues and critical internal mechanical failures.

- * **Justification:** As drone engineers value time efficiency (see Section II), the system should have a way of signalling whether a given issue is caused by some "disposable" part versus a more "critical" piece. This would prevent unnecessary over-corrections while also indicating when greater caution is needed for more significant issues.

- **Goal 2: Identify degraded battery cells.**

- **Objective 2.1:** The system shall measure the Internal Resistance (IR) of individual Lithium Polymer (LiPo) cells with sufficient resolution to distinguish between a healthy (10 - 15 mΩ) [7] and a degraded cell.

- * **Justification:** Internal resistance is the most accurate metric for a battery's "State of Health" (SoH). As LiPo cells age or are damaged by cold Toronto winters, their IR increases, leading to excessive heat and potential fire. Battery cells often fail to deliver consistent current when the IR is not within the healthy range of a LiPo battery [7].

- **Objective 2.2:** The system shall measure the voltage sag detected under a standard stimulated discharge load and distinguish between typical (>90% of nominal voltage) and atypical voltage sag [8].

- * **Justification:** A LiPo battery may provide its full nominal voltage while sitting idle, but it may sag to a dangerously low voltage when the pilot applies full throttle to the motors. This is a primary cause of battery mid-flight failure. Simulating the current draw of a full-throttle drone will validate the battery's performance under the highest motor current demands [8].

- **Goal 3: Adhere to strict volume constraints.**

- **Objective 3.1:** The design shall occupy a volume no larger than a standard handheld multimeter (20 cm x 10 cm x 5 cm) to fit within existing transport cases.

- * **Justification:** Field crews must carry all their equipment into remote locations (forests, rooftops, industrial sites). Space within professional gear bags is extremely limited. A bulky tool will be left in the truck, making it useless for on-site airworthiness verification.

- **Goal 4: Incorporate an intuitive user interface.**

- **Objective 4.1:** The user interface shall require no more than two distinct physical interactions (e.g., plug-in and button press) to initiate a full test sequence [Appendix A, 1.6].

- * **Justification:** High-stress environments like filming sets (see Section I) reduce the cognitive capacity of operators, making complex menus or software interfaces prone to human error.

Studies show that fewer choices correlate with less time spent making decisions, as well as lower cognitive load and error rates [Appendix A, 1.6]. A simple interface ensures that the tool is time-efficient and can be used by any member of the crew regardless of their technical expertise.

- **Goal 5: Ensure functionality in remote and hostile environments.**

- **Objective 5.1:** The system shall operate independently of mains power and internet connectivity.

- * **Justification:** Many GTA filming locations, such as the Scarborough Bluffs or industrial ports, lack reliable Wi-Fi or AC power. A tool that requires a "Cloud Connection" or a wall outlet is a single point of failure for the entire safety protocol.

- **Objective 5.2:** The device shall maintain measurement accuracy in temperatures as low as -15°C [10].

- * **Justification:** Professional filming in Toronto occurs year-round. Standard consumer electronics often suffer from screen lag or sensor drift in the cold. Since cold weather is when LiPo batteries are most likely to fail [Appendix A, 2.13], the diagnostic tool must be reliable even when conditions are at their worst.

- **Goal 6: Create a traceable log of diagnostic results.**

- **Objective 6.1:** The system should be capable of logging results (e.g., "Motor 3: Pass, Battery: Fail") with a timestamp for post-incident analysis and safety audits.

- * **Justification:** In the event of a mid-air failure involving a \$12.5 million liability claim [Appendix A, 2.1], the system must be able to provide clear context and details on the circumstances of failure. A time-stamped log provides the pilot and the production house with proof of due diligence and airworthiness verification. This creates a culture that elevates FPV cinematography from a hobby to a regulated engineering profession.

6 Reference Designs

While the FPV community currently relies on intuition for airworthiness verification, the underlying engineering principles of motor and battery health are well-understood in other industrial sectors. However, existing solutions fail to address the specific time and physical constraints of a professional film set, as well as provide a centralized data log for ease of analysis. The following reference designs support the legitimacy of the problems identified while highlighting the gap that this RFP seeks to fill.

1. Industrial Vibration Monitoring

In the field of industrial maintenance, the health of rotating machinery is routinely monitored using tools such as the Fluke 805 Vibration Meter. These devices use high-sensitivity accelerometers to detect bearing wear and shaft misalignment through Root Mean Square (RMS) velocity measurements [11]. The existence of these high-cost tools validates that vibration is a primary leading indicator of mechanical failure in brushless motors.

The Gap

Industrial meters are designed for stationary machinery with significant mass. They are too bulky for drones as they require manual contact, which is difficult to achieve on small 5-inch drones, and do not provide the rapid, automated pass/fail feedback required within a 30-second production window.



Figure 2: The Fluke 805 Vibration Meter in use on industrial machinery. Note the size of the meter, which makes it unsuitable for the 5-inch carbon fibre arms of an FPV drone.

2. High-Precision Electrochemical Testing

To manage high-performance Lithium Polymer (LiPo) batteries, elite hobbyists occasionally utilize specialized tools like the Wayne Giles ESR Meter. Unlike standard hobbyist voltage checkers, this device uses a precise internal circuit to measure Equivalent Series Resistance (ESR) with milliohm accuracy. This tool validates that Internal Resistance (IR) is a scientifically sound metric for determining a battery's ability to discharge safely under load.

The Gap

The ESR meter is a "bench tool" intended for lab environments. It lacks an integrated housing, is not optimized for rapid "hot-swap" workflows on set, and does not log data for insurance or regulatory compliance. While accurate, its exposed circuitry and lack of a ruggedized housing make it too fragile and infeasible for the harsh, outdoor environments of a film set.



Figure 3: The Wayne Giles ESR Meter, a high-precision tool for battery analysis.

3. Post-Flight Data Analysis (Blackbox)

Modern FPV flight controllers (those equipped with Betaflight firmware) include an onboard "Blackbox" logging feature. This system records gyroscope and PID controller data to an SD card, which can later be analyzed using Fast Fourier Transform (FFT) graphs to identify noise and vibration peaks. This proves that the raw data necessary for airworthiness verification, such as motor noise profiles, is already being generated by the drone's sensors.

The Gap

Blackbox analysis is reactive, not proactive. To use it, a pilot must fly the drone, land, remove the SD card, and review graphs on a laptop. This workflow is impossible to execute in the 30-second window between battery swaps and fails to prevent the "first-flight" failures.

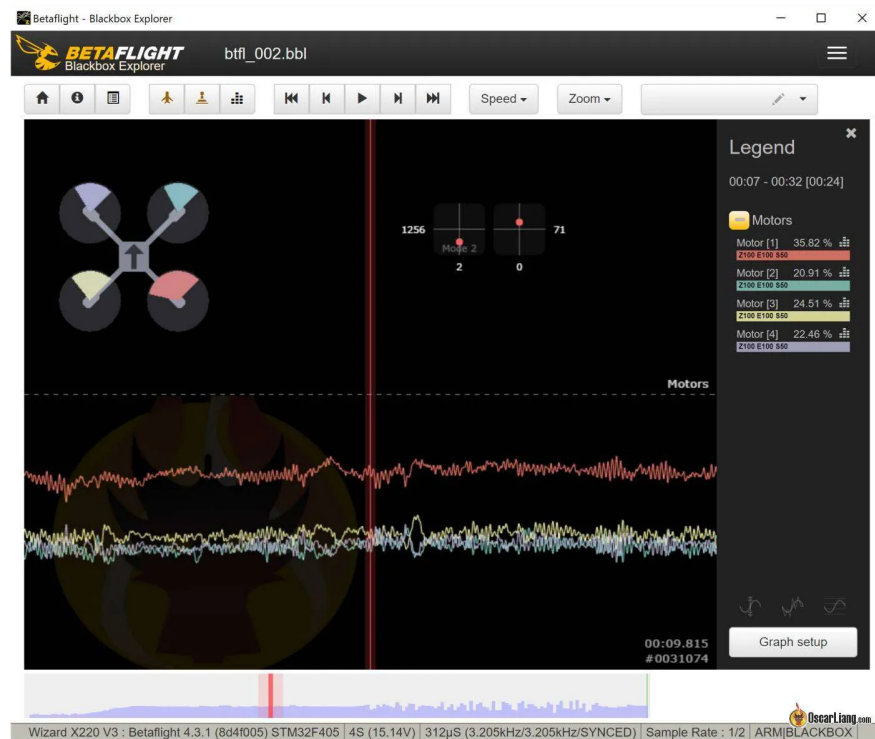


Figure 4: A screenshot of Betaflight Blackbox Explorer showing a graph analysis of motor noise.

4. Smart Batteries

Industry leaders in the drone space, such as DJI, have developed "Intelligent Flight Batteries" that include an integrated Battery Management System (BMS). These chips automatically track cycle counts, health status, and discharge temperatures. The market dominance of smart battery systems proves that automated, objective health tracking is the industry's standard for flight safety.

The Gap

Custom-built professional FPV drones cannot use these systems due to the weight of the integrated circuitry and the requirement for open-source hardware compatibility. The GTA community requires the benefits of a smart system (objective data) applied to "dumb" hardware (lightweight, high-discharge LiPo cells).

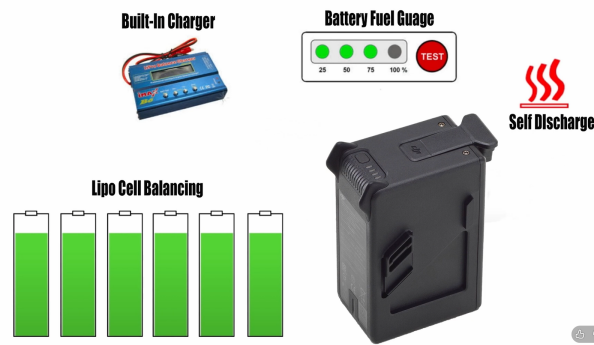


Figure 5: A DJI Intelligent Flight Battery featuring an integrated Battery Management System (BMS).

5. High-Frequency MEMS Sensors

Silicon manufacturers have developed high-bandwidth Micro-Electro-Mechanical Systems (MEMS) sensors, such as the Analog Devices EVAL-CN0532-EBZ, specifically for industrial Condition-Based Monitoring (CbM) [12]. These boards can be used to detect ultrasonic vibration frequencies in large-scale industrial motors before they lead to catastrophic failure. The existence of a dedicated engineering evaluation kit for vibration sensing proves that the subjectivity opportunity with FPV pilots is a solvable problem in the silicon industry. It validates that the technology to "see" internal motor damage through vibration data exists at a hardware level.

The Gap

Evaluation kits are lab development tools intended for electrical engineers, not field-ready equipment for cinematography crews. This board requires an external Data Acquisition (DAQ) system, specialized power supplies, custom housing, and custom software to interpret raw electrical signals into meaningful safety data. The FPV community requires a solution that bridges the gap between this raw "silicon-level" sensing and a "field-level" user interface that provides a simple Go/No-Go status on set.

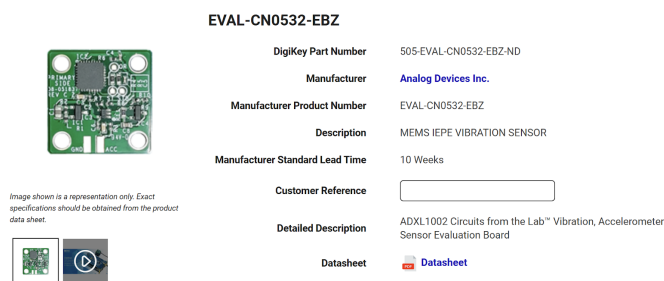


Figure 6: The Analog Devices EVAL-CN0532-EBZ, a MEMS-based vibration sensor evaluation board.

7 Conclusion

The continued reliance on subjective pre-flight checks in the professional FPV drone space presents unacceptable safety and legal vulnerabilities. In the high-pressure environment of a film set, balancing \$12.5 million in potential liability against a constrained 30-second launch window requires more than sensory intuition; it demands objective engineering verification.

This RFP introduced the community of professional FPV drone videography in the GTA, and identified the urgent need for a field-ready diagnostic system which removes subjectivity from the context of pre-flight preparations. The needs are derived directly from primary stakeholder interviews and justified through secondary research. RFP respondents are now invited to design and prototype a solution that respects the outlined constraints and objectives within the six-week project timeline. Ultimately, this RFP frames the engineering opportunity for you, the respondent, to design a metric pre-flight diagnostic solution that respects the stated constraints. Your proposed design will not only enhance safety but also help legitimize the FPV drone videography community in the GTA.

8 References

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A Source Extracts

1. Extracts from References

About this group

A group for Drone flyers. All drones welcome. This group is not about competition, it's about loving the hobby. Whether you're a drone flyer for business or for fun, please be kind to everyone! You are not better than the next person.

RULES

1. Please be polite and respectful to each other.
2. Please do not bully people that are new to the hobby. We are here to help them learn without belittling them.
3. Do not block admins, doing so will result in immediate removal and ban.
4. Absolutely no business advertising unless approved by an admin.
5. Selling your drones or accessories is allowed but price must be listed. Don't do this whole PM ME FOR DETAILS

Thank you, have fun and fly safe!

Steve B. [See less](#)

1.1: About section for the Toronto DJI Drones Facebook group [1]

About us

Everyone is welcome these are free events with geomatics businesses, entrepreneurs, professionals, and students. Mingle with your colleagues and make new friends. You never know when your next contact is going to turn into a business opportunities or job possibility. We hope to see you at the next event. It's totally informal, no speakers to open or close the discussion. Just people like you who are interested in GIS, remote sensing, surveying, and geomatics.

1.2: About section for the Toronto GoGeomatics Canada Meetup group [3]

WHAT WE DO

FPV drones are the latest game-changer in marketing technology, allowing brands to record immersive footage that captures audiences' attention like never before.

We've worked with big name companies in a wide range of industries to deliver standout content that maximizes engagement and leaves a lasting impact.

DANNY NGUYEN

FOUNDER

Danny got into FPV drone flying nearly 10 years ago, captivated by the sensation of truly flying. The immersive perspective hooked him instantly, blending precision, speed, and creativity that continually pushed his boundaries as a FPV drone pilot and filmmaker. Over the decade, he honed his skills to capture unique, dynamic perspectives, bringing stories to life in ways traditional cameras couldn't.

1.3: About sections for D.Vision Aerials and Danny Nguyen [4]

For standard short films, production costs typically range between \$700-1,500 per minute of screen time. A 10-minute project generally requires between \$7,000-15,000. However, the median budget stands at \$3,000, indicating that many filmmakers complete their projects with more modest resources.

1.4: Excerpts from budgeting for short films, specifying the cost per minute of filming [5]

3. Drone Propeller Damage and Imbalance

Propellers are directly responsible for lift and stability.

- Cracks and bends: After a hard landing or collision, tiny cracks or warps may go unnoticed—until a blade snaps mid-flight and you lose thrust in an instant.

1.5: Excerpt from ligpower stating that tiny cracks may go unnoticed until too late [6].

This seminal research established the principle originally known as Hick's Law (Hick, 1952), and now the Hick-Hyman Law, (Hyman, 1953). It can be summarized as: the time it takes for a person to make a decision increases logarithmically (note: $\log_2$) with the number of choices they have. Each additional option adds complexity to the decision-making process, requiring more time to evaluate and choose. Put simply, the more choices people have, the longer it will take them to pick one.

Reducing the number of choices can have profound effects on cognitive load and error rates. For example, decreasing choices (and therefore the amount of time it takes to make a choice) can lower the mental effort required to process information, which [reduces cognitive strain and helps individuals make decisions with less mental fatigue](#). When people have an easier time making decisions, they can be more efficient and accurate, and the experience can be more satisfying!

1.6: Excerpts from M. Moore detailing the impacts of the number of choices an individual is given on the time-efficiency and cognitive load [9].

2. Interview Excerpts – Danny Nguyen, D.Vision Aerials (26 Jan. 2026)

*All information from quotes is relevant information. "... " is a placeholder for irrelevant information

[2.1] On the difference between a pro and a hobbyist:

"You wouldn't want a guy who just picked this up as a hobby to do this. You want people like fully licensed, fully insured... because if anything goes wrong, it comes back to your company and if you get sued, you better have everything covered. ... We're covered for like \$12.5 million in liability."

[2.2] On the sensory experience:

"It's like it's an out-of-body experience. ... I'll be like moving my body thinking I'm moving the drone, but I'm not. ... I'm like a kilometer away. But in my mind I'm like, yo, I feel like I'm in Star Wars."

[2.3] On "imposter syndrome":

"I'm watching this helicopter do some like crazy like shots... The red camera is probably worth like \$80,000 attached to the helicopter... And I'm like, in my head, I'm like, wow, like I have imposter syndrome. I shouldn't be here. ... [But the producer said] 'Dude, I don't know what you just cooked up, but dude, that was some of the craziest shots I've ever seen in my entire life.'"

[2.4] On the pilots' attitude on set:

"Sometimes we go like, wow, is this like, is this even real life? ... The guy's just like, his eyes are just like so big. He's just like, what? That was sick. ... You guys have no fear. And I think that "no fear" part comes from really tuning in our skills and knowing that we got it."

[2.5] On the subjectivity of motor checks:

"Well, kinda. Like if a motor is going bad, you know, obviously just through audio hearing, you hear this one just doesn't sound right. 'Cause we fly this one so often. We know what it should sound like. If we hear like a spider, a little Creek, we're like, OK. Let's not fly this drone. Let's look into it. Maybe take off the motor, look into it. Maybe there's a rock stuck in it. Maybe it's rusted, or maybe the ESC or the flight controller is pushing too much power to one side, and it's like it's making it spin out and all that stuff. So then we start diagnosing and we basically work from I guess the top down."

[2.6] On battery quality control:

"I have batteries that I bought maybe three years ago that are still good. I have batteries that I bought six months ago that are already dying, and the quality control is really hard to tell because these batteries are just being pumped out."

[2.7] On RF interference:

"Oh, someone just used an Amazon Alexa that just triggered a 2.8 gigahertz channel and just blew us out of the air. ... We found out in the logbook that looks like some spike hit up... I was like, wow, who knew?"

[2.8] On data paranoia:

"What's your biggest fear regarding the footage...? Did we press record? Oh my God, dude. ... Like, that's like, Oh my God, pain point right there, bro. Did we press record? ... Or is it corrupted? ... Oh ****, this SD card is cooked like it's been ran through the bushes and back."

[2.9] On the "hot swap" voltage problem:

"The pain point for me would be... we need this drone running for 8 hours straight... [If] you plug a battery in... The voltage is going to spike and then like corrupt or like destroy the other battery. ... So we've been trying to figure that out for the longest time."

[2.10] On weight constraints:

"Weight is the biggest factor in the drone... especially on like a bigger drone you can get away with like an extra 2-3 hundred grams, but on something like this tiny, if you add like 10 grams, it's not gonna fly the right way."

[2.11] On setup time:

"They go, Danny, it's your turn to shoot. How long do you need? 5 minutes? No, we're like 30 seconds. Just plug in the battery, strap it in, put my goggles on, put my controller on, and then boom, we're off."

[2.12] On durability:

"We ended up looking into like a two-tiered drone... because it's now it's all carbon fiber versus like before it was all plastic, so you can take a bigger beating."

[2.13] On cold weather performance:

"If I don't keep the batteries warm... I probably get 60% life off the battery. ... If it's a cold day... I'll probably only get like 2 ½ minutes of flight and that's just because of how cold affects conductivity."

[2.14] On the complexity of regulations:

"Within Canada, you need to have a drone license if you want to operate in certain areas of the city... if you go downtown, you're surrounded by Billy Bishop. ... You have to understand the risk that you're taking, especially if you're interfering with like air traffic... understanding NOTAMs and all these other things."

[2.15] On the changing rules:

"They just changed it recently last year... before, under 250 grams you didn't need a license... now... you still should get a license and still have to apply for permits."

[2.16] On the drone speed:

"...this thing goes almost like 180 kilometers an hour."

B Photos from the Interview

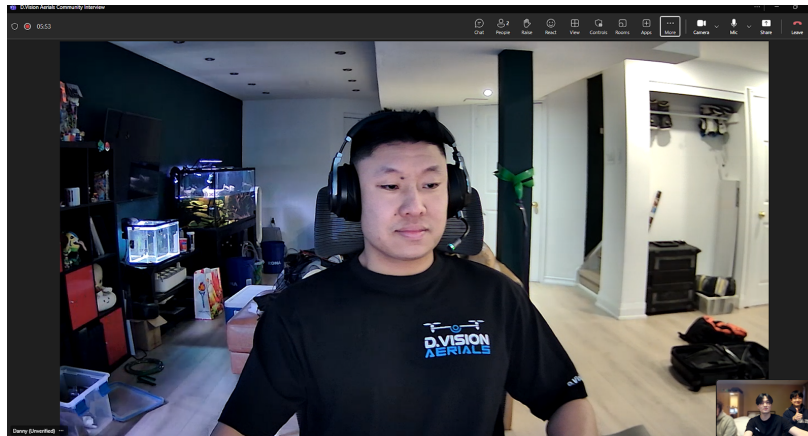


Figure 7: Nguyen and interviewers



Figure 8: Nguyen displaying various drone sizes used by D.Vision Aerials



Figure 9: Nguyen showing a 'Capacity Controller' used for testing batteries



Figure 10: Nguyen presenting a bundle of dead batteries